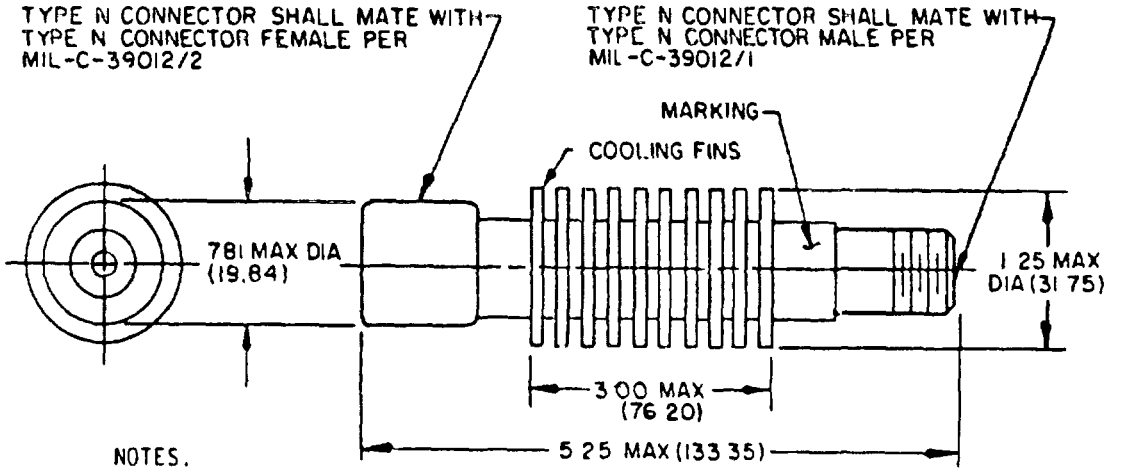
MILITARY SPECIFICATION SHEET

ATTENUATORS, FIXED COAXIAL LINE (SERIES N),

FREQUENCY RANGE DC TO 18 GHz, CLASS III, MEDIUM POWER

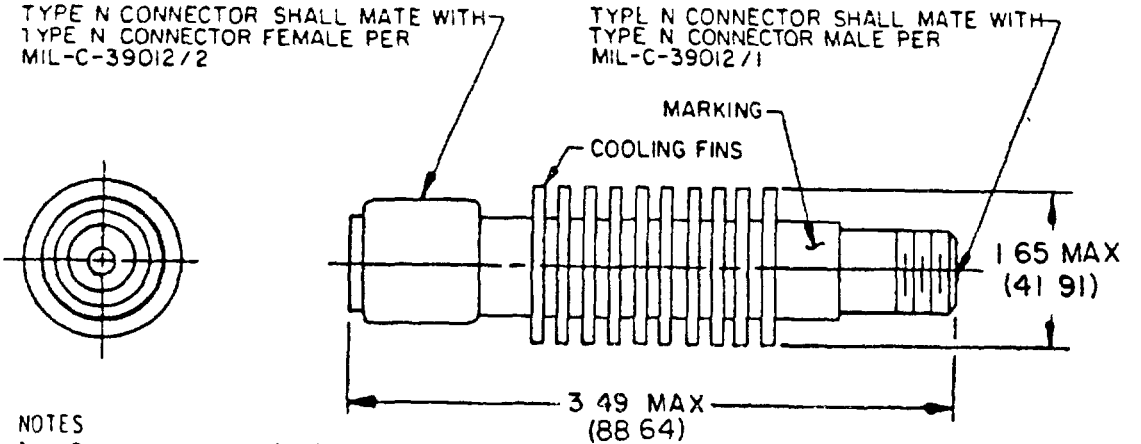
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the attenuators described herein shall consist of this specification and the latest issue of MIL-A-3933.



NOTES.
1 Dimensions are in inches.
2 Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1 Dimensions and configuration for Part number M3933/10-1



NOTES
1 Dimensions are in inches.
2 Metric equivalents are given for general information only.
3 Metric equivalents are in parentheses

FIGURE 2 Dimensions and configuration for Part number M3933/10-2 thru /10-22.

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ENGINEERING DATA:

Class: III.

Operating frequency range: DC to 18 gigahertz (GHz) (see table I).

Test frequencies for VSWR and attenuation measurements.

DC, 1, 2, 3, and 4 GHz for dash numbers 01 thru 05.
DC, 2, 4, 6, and 8 GHz for dash numbers 06 thru 09.
DC, 4, 6, 8, and 11 GHz for dash numbers 10 thru 14.
DC, 4, 8, 12.4 and 18 GHz for dash numbers 15 thru 22.

Basic test frequency:

2 GHz for dash numbers 01 thru 05.
4 GHz for dash numbers 06 thru 09.
6 GHz for dash numbers 10 thru 14.
8 GHz for dash numbers 15 thru 22.

Operating temperature range.

M3933/10-1: -55°C to +85°C.

M3933/10-2 thru 22: -55°C to +125°C.

Maximum weight: 6 ounces.

REQUIREMENTS.

Dimensions and configuration See figures 1 and 2. Cooling fins shall be circular.

Nominal impedance 50 ohms

Attenuation See table I.

Power. See table I.

VSWR: See table I.

Part number M3933/10- (and dash number from table I).

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TABLE I Electrical characteristics.

Dash number	Attenuation dB								Maximum power input		VSWR maximum 1/					
	Nominal	Deviation														
		DC	DC to 4.0 GHz	DC to 6 GHz	DC to 8 GHz	DC to 11 GHz	DC to 12 GHz	DC to 18.0 GHz	AV (W) @ 25°C (continuously)	Peak 2/ (kW)	dc to 4.0 GHz	4.0 to 6 GHz	4.0 to 8.0 GHz	6.0 to 11 GHz	8.0 to 12.0 GHz	12.0 to 18.0 GHz
01 N S	6	±0.1	±0.2	---	---	---		---	15	3 2/	1.30 1	---	---	---	---	---
02 N S	3	±0.3	±0.5	---	---	---		---	20 4/	1 3/	1.15 1	---	---	---	---	---
03 N S	10	±0.3	±0.5	---	---	---		---	20 4/	1 3/	1.15 1	---	---	---	---	---
04 N S	20	±0.3	±0.5	---	---	---		---	20 4/	1 3/	1.15·1	---	---	---	---	---
05 N S	30	±0.4	±0.75	---	---	---		---	20 4/	1 3/	1.15 1	---	---	---	---	---
06 N S	3	±0.2	±0.3	---	±0.6	---		---	25 5/	5 2/	1.20 1	---	1.30 1	---	---	---
07 N S	6	±0.2	±0.3	---	±0.6	---		---	25 5/	5 2/	1.20 1	---	1.30.1	---	---	---
08 N S	10	±0.2	±0.3	---	±0.6	---		---	25 5/	5 2/	1.20 1	---	1.30 1	---	---	---
09 N S	20	±0.2	±0.3	---	±0.6	---		---	25 5/	5 2/	1.20 1	---	1.30 1	---	---	---
10 N S	30	±0.4	±0.6		±1.0	---		---	25 5/	5 2/	1.20 1	---	1.30 1	---	---	---
11 N S	3	±0.15	±0.35	±0.65	---	±0.75		---	20	1 2/	1.15 1	1.20 1	---	1.30 1	---	---
12 N S	6	±0.15	±0.35	±0.65	---	±0.75		---	20	1 2/	1.15·1	1.20 1	---	1.30.1	---	---
13 N S	10	±0.15	±0.35	±0.65	---	±0.75		---	20	1 2/	1.15 1	1.20 1	---	1.30 1	---	---
14 N S	20	±0.15	±0.35	±0.65	---	±0.75		---	20	1 2/	1.15.1	1.20 1	---	1.30:1	---	---
15 N S	30	±0.15	±0.35	±0.65	---	±0.75		---	20	1 2/	1.15.1	1.20:1	---	1.35 1	---	---

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB								Maximum power input		VSWR maximum 1/					
	Nominal	Deviation														
		DC	DC to 4.0 GHz	DC to 6 GHz	DC to 8 GHz	DC to 11 GHz	DC to 12 GHz	DC to 18.0 GHz	AV (W) @ 25°C (continuously)	Peak 2/ (kW)	dc to 4.0 GHz	4.0 to 6 GHz	4.0 to 8.0 GHz	6.0 to 11 GHz	8.0 to 12.0 GHz	12.0 to 18.0 GHz
16 N S	3	±0.2	---	---	---	---		±0.3	20 5/	1 2/	1.15:1	---	1.20:1	---	1.25.1	1.40:1
17 N S	6	±0.2	---	---	---	---		±0.3	10 5/	1 2/	1.15.1	---	1.20:1	---	1.25·1	1.40:1
18 N S	10	±0.3	---	---	---	---		±0.5	10 5/	2 2/	1.15·1	---	1.20·1	---	1.25:1	1.40 1
19 N S	20	±0.3	---	---	---	---		±0.5	10 5/	2 2/	1.15 1	---	1.20·1	---	1.25.1	1.40 1
20 N S	30	±0.6	---	---	---	---		±1.0	10 5/	2 2/	1.15·1	---	1.20:1	---	1.25 1	1.40 1
21 N S	40	±0.6	---	---	---	---		±1.0	10 5/	2 2/	1.15 1	---	1.20.1	---	1.25 1	1.40 1
22 N S	50	±1.0	---	---	---	---		±1 35	10 5/	2 2/	1.15 1	---	1.20 1	---	1.25:1	1.40.1

1/ VSWR value is for both ends.

2/ Peak power for a duty cycle of 5×10^{-4} maximum duration 5 microseconds.

3/ Peak power for a duty cycle of 1×10^{-3}.

4/ Derate linearly from 20 watts at 25°C to 10 percent at 125°C

5/ Derate linearly from the stated voltage at 25°C to 0 watts at 125°C.

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Note. Attenuation covered by this specification sheet are substitutable for the following manufacturer's or user's part numbers specified in table II. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number.

TABLE II. Substitution information.

Military part number M3933/10-	Manufacturer's code	Manufacturer's part number	NSN
01	80058	CN1301	5985-00-445-6953
01	00929	AD06N	5985-00-445-6953
02	34078	303-3	5985-01 153-2658
02	05869	721768 1	---
02	99899	765-3	5985-00-233-7291
03	96214	539071-3	5985-00-104-8513
03	99899	766-10	5985-00-104-8513
03	00929	AD10N	5985-00-445-6952
03	05869	721768-3	---
04	99899	766-20	5985-00-764-2531
04	05869	721768-4	---
04	00929	AD20N	5985-00 475 4234
05	05869	721768-5	5985-00-145-9720
05	96214	539071-5	5985-00-145-9720
05	99899	766-30	5985-00-145-9718
06	93459	33-3-34	5985-01-104-7533
07	93459	33-6-34	---
08	93459	33-10-34	---
09	93459	33-20-34	---
10	18203	8430N	5985-00-233-2208
10	93459	33-30-34	---
11	99899	768-3	---
12	99899	768-6	5985-01-078-6821
13	99899	768-10	5985-00-241-8599
14	99899	768-20	5985-00-256-8449

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TABLE II. Substitution information - Continued.

Military part number M3933/10-	Manufacturer's code	Manufacturer's part number	NSN
15	99899	768-30	5985-01-118-4416
15	99899	768-30	5985-00-233-4626
16	93459	23-3-34	---
17	93459	22-6	5985-00-967-2725
17	93459	23-6-34	---
18	93459	22-10	5985-01-054-5561
18	93459	23-10-34	---
19	93459	23-20-34	---
20	93459	22-30	5985-01-071-0510
20	93459	23-30-34	---
21	93459	22-40	5985-01-060-4751
21	93459	23-40-34	---
22	80063	SM-C-920679	---
22	93459	23-50-34	---

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 11, 99
DLA - ES

User activities.
Army - SM
Navy - MC, SH, AS

Agent:
DLA - ES

Preparing activity
Navy - EC

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	☐ MANUFACTURER
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b Recommended Wording

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